# Bitcoin and technology

## 2018-19:

Section A

Question 1 – Fundamentals [25 Marks] A) The release of the Bitcoin Network in 2008/2009 through the pseudonym of Satoshi Namakoto heralded the advent of Blockchain technologies. This technology has come about as a result of a combination of previous works in the area of Cryptography, Consensus Mechanisms, e-Currencies and electronic contracts. Highlight 3 key contributors and discuss the technologies they have created which have led to the advent of the Bitcoin Network, additional marks provided for accurate citations of white papers [15 Marks]

ANSWER: **Key contributors and technologies leading to the advent of the Bitcoin Network**

Bitcoin did not emerge in isolation. Its release in 2008/2009 under the pseudonym **Satoshi Nakamoto** was the result of several decades of foundational research in cryptography, distributed systems, and digital money. Below are **three key contributors** whose work directly enabled the creation of the Bitcoin network, supported by references to their original white papers.

### 1. Whitfield Diffie & Martin Hellman – Public Key Cryptography

**Contribution**
In 1976, **Whitfield Diffie** and **Martin Hellman** published *"New Directions in Cryptography"*, introducing **public key cryptography**.

**Technology introduced**

- Asymmetric cryptographic key pairs (public/private keys)
- Secure communication without pre-shared secrets
- Digital signatures for authentication and non-repudiation

**Impact on Bitcoin**
Bitcoin relies on elliptic curve public key cryptography (ECDSA) to:

- Prove ownership of funds
- Sign transactions securely
- Eliminate the need for trusted intermediaries

Without public key cryptography, Bitcoin's trustless ownership model would not be possible.

## 2. Adam Back – Hashcash and Proof of Work

### Contribution
In 1997, **Adam Back** proposed *Hashcash*, a system designed to combat email spam.

### Technology introduced

- **Proof of Work (PoW)** using cryptographic hash puzzles
- Computational cost as a deterrent against abuse
- Verifiable but expensive-to-produce work

### Impact on Bitcoin
Satoshi directly cites Hashcash in the Bitcoin white paper. Bitcoin repurposes Proof of Work to:

- Secure the blockchain
- Achieve decentralised consensus
- Prevent double-spending without a central authority

Hashcash transformed energy expenditure into a **security mechanism**, forming the backbone of Bitcoin mining.

## 3. Nick Szabo – Smart Contracts and Bit Gold

### Contribution
**Nick Szabo** introduced the concept of **smart contracts** in the 1990s and later proposed *Bit Gold* (1998), a decentralised digital currency design.

### Technology introduced

- Self-executing contracts enforced by code
- Trust minimisation through automation
- Decentralised digital scarcity via cryptographic puzzles

### Impact on Bitcoin
While Bit Gold was never implemented, it introduced:

- Chained cryptographic proofs
- Decentralised currency without a central issuer
- Concepts later refined and realised in Bitcoin

Bitcoin can be viewed as the first **successful implementation** of Szabo's theoretical framework.

### Critical appraisal

These contributors were selected because their work maps directly onto Bitcoin's core architecture:

- Diffie and Hellman enabled **cryptographic ownership**
- Adam Back provided **consensus and security**
- Nick Szabo laid the groundwork for **digital money and smart contracts**

Satoshi Nakamoto's innovation was not inventing these ideas, but **combining them into a single, self-sustaining system** governed by mathematics, incentives, and code rather than institutions.

**Referenced white papers**

- Diffie & Hellman – *New Directions in Cryptography* (1976)
- Adam Back – *Hashcash: A Denial of Service Counter-Measure* (1997)
- Nick Szabo – *Smart Contracts* (1994), *Bit Gold* (1998)
- Satoshi Nakamoto – *Bitcoin: A Peer-to-Peer Electronic Cash System* (2008)

B) Critically Discuss and Evaluate the below consensus mechanisms, within your discussion ensure to include in what networks that these mechanisms are found, purpose of implementing such consensus mechanisms and how they differ from one another.

• Proof of Work • Proof of Burn • Proof of Stake. • Proof of Elapsed Time [10 Marks]

ANSWE: All consensus mechanisms solve the same core problem: **how a decentralised network agrees on a single version of truth without trusting any one participant**. Where they differ is in *cost, security assumptions, and incentives*. Below is a critical evaluation of the four mechanisms listed, including where they are used, why they exist, and how they differ.

**Proof of Work (PoW)**

**Where it is used**
PoW is most notably used in **Bitcoin** and, historically, in Ethereum prior to its transition to Proof of Stake.

**Purpose**
PoW secures the network by requiring miners to solve cryptographic hash puzzles. The computational difficulty ensures that rewriting the blockchain is economically infeasible.

**Evaluation**

- **Strengths**: Extremely secure, battle-tested, and resistant to Sybil attacks. Security is tied directly to real-world energy and hardware costs.
- **Weaknesses**: High energy consumption, increasing centralisation due to mining pools, and limited scalability.

**Critical insight**
PoW converts energy into trust, which is effective but environmentally and economically expensive.

**Proof of Burn (PoB)**

**Where it is used**
PoB has been implemented in networks such as **Slimcoin** and discussed in experimental blockchain designs.

**Purpose**
Instead of expending energy, participants "burn" coins by sending them to an irretrievable address. The act proves commitment and grants mining or validation rights.

**Evaluation**

- **Strengths**: Reduces ongoing energy consumption and discourages short-term attacks.
- **Weaknesses**: Permanent destruction of value can be economically wasteful and discouraging for participants.

**Critical insight**
PoB replaces energy waste with capital destruction, which shifts rather than eliminates inefficiency.

**Proof of Stake (PoS)**

**Where it is used**
PoS is used by modern networks such as **Ethereum**, Cardano, and Solana.

**Purpose**
Validators are selected based on the amount of cryptocurrency they stake. Misbehaviour results in slashing of staked assets.

**Evaluation**

- **Strengths**: Drastically lower energy usage, faster finality, and improved scalability.
- **Weaknesses**: Wealth concentration risks and complex governance mechanisms.

**Critical insight**
PoS secures the network through economic penalties rather than physical resource expenditure, trading energy cost for financial risk.

**Proof of Elapsed Time (PoET)**

**Where it is used**
PoET is primarily used in permissioned blockchain networks such as **Hyperledger Sawtooth**.

**Purpose**
Participants wait for a randomly assigned time before proposing a block. The node with the shortest wait time wins.

**Evaluation**

- **Strengths**: Energy-efficient and suitable for enterprise environments.
- **Weaknesses**: Relies on trusted execution environments (e.g. Intel SGX), reducing decentralisation.

**Critical insight**
PoET sacrifices trustlessness for efficiency, making it unsuitable for open, permissionless networks.

**Comparative Evaluation**

| Mechanism | Security Basis | Energy Use | Decentralisation | Typical Use |
|---|---|---|---|---|
| PoW | Computational work | Very High | High (but pool-driven) | Public blockchains |
| PoB | Economic sacrifice | Low | Moderate | Experimental |
| PoS | Financial stake | Very Low | Variable | Modern public chains |
| PoET | Trusted hardware | Very Low | Low | Permissioned chains |

## Conclusion

What this really means is that **no consensus mechanism is universally "best"**.

- PoW prioritises security over efficiency.
- PoS optimises sustainability and scalability.
- PoB experiments with economic commitment.
- PoET suits controlled enterprise environments.

Each mechanism reflects a different trade-off between trust, cost, and decentralisation, chosen based on the network's goals and threat model.

Section B - Answer 3 out of the below 4 Questions

Question 2 – Foundations [25 Marks]

A) The Blockchain is comprised of a multitude of individual blocks linked together to form a chain. Explain in detail the composition of each individual block, highlighting the key components of the block header data [4 Components] and discuss the

importance of each component within the header towards the functionality of the chain. [10 Marks]

ANSWER: A blockchain is a **linked list of blocks**, and what actually *links* those blocks together is the **block header**. The header is small in size but critical in function. It is the part of the block that is hashed, validated, and agreed upon by the network.

A typical blockchain block (for example in **Bitcoin**) consists of:

- **Block Header** (metadata used for consensus)
- **Block Body** (transactions)

Your focus here is the **block header**, which contains the core data that secures the chain.

# Key Block Header Components (4 components)

## 1. Previous Block Hash

**What it is**
A cryptographic hash of the previous block's header.

**Why it matters**

- Creates the **chain** in blockchain
- Any change to a previous block alters its hash, breaking all subsequent links
- Provides immutability and tamper resistance

**Importance**
This is the mechanism that makes historical data effectively unchangeable without re-mining the entire chain.

## 2. Merkle Root

**What it is**
A single hash representing all transactions in the block, generated from a Merkle Tree.

**Why it matters**

- Efficiently summarises potentially thousands of transactions
- Enables lightweight clients to verify transactions without downloading full blocks

**Importance**
If even one transaction is altered, the Merkle root changes, invalidating the block. This guarantees **transaction integrity**.

### 3. Timestamp

**What it is**
The approximate time at which the block was created.

**Why it matters**

- Helps order blocks chronologically
- Supports difficulty adjustment mechanisms
- Prevents certain replay and timing-based attacks

**Importance**
Timestamps keep the blockchain progressing forward in time and help maintain consensus on block order.

### 4. Nonce (with Difficulty Target)

**What it is**
A variable number miners change repeatedly to find a valid hash under the network's difficulty target.

**Why it matters**

- Core component of **Proof of Work**
- Introduces computational difficulty
- Makes block creation costly and verifiable

**Importance**
The nonce transforms hashing into a security mechanism. It ensures that adding a block requires real-world effort, protecting the network from spam and attacks.

# Why the block header is critical to blockchain functionality

What this really means is that the block header:

- Secures the blockchain through cryptography
- Links blocks immutably
- Enables decentralised consensus
- Prevents tampering and double spending

Without the block header, a blockchain would simply be an unordered database. With it, the system becomes **trustless, verifiable, and secure**, allowing independent nodes to agree on a single shared history without relying on any central authority.

B) Define the four fundamental concepts, which you believe make blockchain possible, and how these concepts interact and combine to enable blockchain as a disruptive technology. You must also critically appraise why they are the most fundamental concepts. [15 marks]

ANSWER: **Four fundamental concepts that make blockchain possible (15 marks)**

Here's the thing. Blockchain is not a single invention. It is a **carefully engineered combination of existing concepts** that, when combined, produce something genuinely disruptive. Many technologies contribute to blockchain, but the four below are the most fundamental because **if you remove any one of them, blockchain stops working as intended**.

# 1. Cryptography (Hashing and Digital Signatures)

**Definition**
Cryptography provides the mathematical foundations for security, integrity, and ownership in blockchain systems. This includes:

- Cryptographic hash functions
- Public–private key cryptography
- Digital signatures

**How it enables blockchain**

- Hashing ensures data integrity and immutability. Any change to data produces a completely different hash.
- Digital signatures prove ownership of assets and authorise transactions without revealing private keys.
- Public key cryptography removes the need for trusted intermediaries.

**Critical appraisal**
Cryptography is fundamental because blockchain trust is **mathematical, not institutional**. Without cryptography, there is no secure ownership, no immutability, and no trustless verification. However, cryptography alone does not solve agreement or coordination problems, which is why it must be combined with other concepts.

# 2. Distributed Ledger Technology (DLT)

**Definition**
A distributed ledger is a database that is **replicated and synchronised across many independent nodes**, rather than controlled by a central authority.

**How it enables blockchain**

- Every participant holds a copy of the ledger.

- There is no single point of failure.
- Transparency and auditability are built into the system.

**Critical appraisal**
DLT is essential because it removes central control and censorship. A cryptographically secure ledger stored on a single server would not be disruptive. The disruption comes from **shared ownership of state across a network of untrusted participants**. On its own, however, a distributed ledger still needs a way to agree on updates, which leads directly to consensus.

# 3. Consensus Mechanisms

**Definition**
Consensus mechanisms are protocols that allow decentralised nodes to **agree on the valid state of the ledger**, even in the presence of malicious actors.

**How it enables blockchain**

- Prevents double spending
- Ensures only valid transactions are added
- Aligns participants through incentives and penalties

**Critical appraisal**
Consensus is what allows blockchain to function without a central authority. Traditional distributed systems assume trusted participants; blockchain explicitly assumes the opposite. While consensus mechanisms vary, the underlying concept of decentralised agreement is non-negotiable. Without consensus, a distributed ledger quickly fragments into conflicting versions of history.

# 4. Economic Incentives and Game Theory

**Definition**
Blockchain systems use economic rewards and penalties to **align rational self-interest with honest network behaviour**.

**How it enables blockchain**

- Validators or miners are rewarded for following the rules.
- Dishonest behaviour is penalised financially.
- Network security becomes economically self-enforcing.

A) Critically assess how the use of Hashing Algorithms contribute to the functionality of the blockchain network, citing key contributions and implementations within the network. [8 Marks]

## ANSWER: Role of hashing algorithms in blockchain (point-wise, short answer)

- **Data integrity**
  Hashing ensures that any change to block or transaction data produces a different hash, making tampering immediately detectable.
- **Block linking (immutability)**
  Each block stores the hash of the previous block, cryptographically chaining blocks together and securing the entire history.
- **Transaction verification**
  Hashing is used in Merkle Trees to efficiently verify transactions without processing the entire block.
- **Consensus support**
  In Proof of Work systems, hashing provides the computational puzzle that secures the network and prevents double spending.
- **Security and identity**
  Blockchain addresses are generated from hashed public keys, improving privacy and reducing attack exposure.

Together, these points show how hashing enables a secure, tamper-resistant blockchain network.

B) The Cypherpunk manifesto initially posed by Tim May and Eric Hughes highlights some key principles that should be understood concerning the rights of individuals to privacy online. Explain these 4 principles and the technologies implemented to address them. [10 Marks]

## ANSWER: Cypherpunk Manifesto: principles and supporting technologies (10 marks)

The Cypherpunk Manifesto, articulated by **Eric Hughes** and influenced by **Timothy C. May**, argues that privacy is a fundamental right and that **technology, not legislation, is the only reliable way to protect it**. The manifesto outlines several core principles; four key ones are explained below, along with the technologies that implement them.

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## 1. Privacy as a fundamental right

- **Principle**: Individuals must be able to control how and when their personal information is revealed.
- **Technology used**:
  - Public key cryptography
  - Encryption (asymmetric and symmetric)

These tools allow secure communication without exposing identity or sensitive data.

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## 2. Privacy through anonymity and pseudonymity

- **Principle**: True privacy requires the ability to interact without revealing real-world identity.
- **Technology used**:
  - Cryptographic key pairs
  - Blockchain wallet addresses
  - Mixers and privacy-preserving protocols

This enables participation in digital systems without central identity authorities.

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## 3. Trust minimisation

- **Principle**: Systems should not require users to trust governments, corporations, or intermediaries.
- **Technology used**:
  - Decentralised networks
  - Blockchain and distributed ledgers
  - Peer-to-peer protocols

Trust is replaced with mathematical verification and open-source code.

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## 4. Code as enforcement (crypto over law)

- **Principle**: Privacy must be enforced by cryptographic systems rather than legal promises.
- **Technology used**:
  - End-to-end encryption
  - Smart contracts
  - Cryptographic protocols

Once deployed, these systems enforce privacy automatically and consistently.

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## Conclusion (short)

These principles are fundamental because they shift privacy protection from **policy to protocol**. The Cypherpunk movement directly influenced modern privacy technologies and laid the ideological and technical groundwork for cryptocurrencies and blockchain systems, where privacy is preserved through cryptography and decentralisation rather than trust in institutions.

C) Digital Signatures now play a key role in everyday business operations within the FinTech and ICT domains. Appraise and discuss the key attributes, which digital signatures provide. [7 Marks]

## ANSWER: Key attributes provided by digital signatures (7 marks, point-wise)

- **Authentication**
  Digital signatures verify the identity of the sender, proving that a message or transaction was created by the legitimate private key holder.
- **Integrity**
  Any alteration to the signed data invalidates the signature, ensuring the content has not been tampered with.
- **Non-repudiation**
  The signer cannot deny having signed the message or transaction, as only their private key could have produced the signature.
- **Trust without intermediaries**
  Digital signatures remove the need for third-party verification by enabling cryptographic proof of authenticity.
- **Legal and regulatory acceptance**
  Digital signatures are recognised in many jurisdictions as legally binding, supporting secure digital contracts and compliance.
- **Automation and efficiency**
  They enable automated processing of transactions and contracts, reducing time, cost, and human error.
- **Security in decentralised systems**
  In blockchain and FinTech systems, digital signatures secure transactions and enforce ownership without central control.

Together, these attributes make digital signatures a foundational security mechanism in modern FinTech and ICT systems.

Question 4 – Crypto Economics[25 Marks]

A) Explain how Bitcoin are created, within your response highlight and detail key functions and standardizations within the process. [5 Marks]

## ANSWER: A) How Bitcoins are created (short, point-wise – 5 marks)

- **Mining process**
  New Bitcoins are created through **mining**, where miners compete to add a new block to the **Bitcoin**.
- **Proof of Work**
  Miners perform repeated hashing to find a block hash below the network's difficulty target, proving computational work.
- **Block reward (coinbase transaction)**
  The successful miner receives newly minted Bitcoins via a **coinbase transaction**, which has no inputs and creates new coins.
- **Difficulty adjustment**
  The network automatically adjusts mining difficulty every 2016 blocks to maintain an average block time of approximately 10 minutes.
- **Controlled supply standardisation**
  Bitcoin's issuance follows a fixed monetary policy: a maximum supply of 21 million coins, with block rewards halving approximately every four years.

Together, these mechanisms ensure that Bitcoin creation is **decentralised, predictable, and resistant to inflation**, forming the economic foundation of the network.

B) Discuss in detail how mining and reward mechanisms play a key role within the Bitcoin network, specifically highlight the roles the miners play in securing the network. [10 Marks]

## ANSWER: Role of mining and reward mechanisms in securing the Bitcoin network (10 marks)

Mining and reward mechanisms are central to how the **Bitcoin** operates securely without a central authority. They align economic incentives with honest behaviour and protect the network from attacks.

- **Transaction validation**
  Miners verify that transactions follow protocol rules, including correct signatures and sufficient balances, before inclusion in a block.
- **Block creation**
  Miners assemble valid transactions into blocks and attempt to add them to the blockchain through Proof of Work.

- **Proof of Work security**
  Mining requires substantial computational effort, making attacks such as double spending economically unfeasible.
- **Network consensus**
  The longest valid chain (most accumulated work) is accepted as the canonical ledger, allowing decentralised agreement.
- **Immutability enforcement**
  Altering past transactions would require re-mining all subsequent blocks, providing strong resistance to history rewriting.
- **Block rewards**
  Miners are incentivised through newly minted Bitcoins and transaction fees, motivating participation and honesty.
- **Economic deterrence**
  Any malicious behaviour risks wasted energy and lost opportunity costs, discouraging attacks.
- **Decentralisation of trust**
  Mining distributes control across independent participants rather than a central authority.
- **Self-regulating security**
  As Bitcoin's value increases, mining competition intensifies, strengthening network security.
- **System sustainability**
  Over time, transaction fees are expected to replace block rewards, maintaining miner incentives.

Together, mining and reward mechanisms transform economic self-interest into a **robust security model**, allowing Bitcoin to function as a decentralised, trustless monetary system.

C) Provide critical insight into the role of crypto currencies on existing financial markets their advantages and disadvantages and the potential disruptive nature of these new currencies. [10 Marks]

## ANSWER: Role of cryptocurrencies in financial markets: advantages, disadvantages, and disruption (10 marks)

Cryptocurrencies have introduced a parallel financial system that challenges many assumptions of traditional finance. Their impact is best understood by weighing both benefits and limitations.

### *Advantages*

- **Disintermediation**
  Cryptocurrencies enable peer-to-peer value transfer without banks or clearing houses, reducing costs and settlement times.
- **24/7 global markets**
  Crypto markets operate continuously across borders, unlike traditional exchanges with limited trading hours.

- **Financial inclusion**
  Anyone with internet access can participate, benefiting unbanked or underbanked populations.
- **Transparency**
  Public blockchains provide open, auditable transaction records, improving trust and accountability.

### *Disadvantages*

- **High volatility**
  Price instability limits cryptocurrencies' effectiveness as a stable medium of exchange.
- **Regulatory uncertainty**
  Inconsistent regulations create risk for investors and institutions.
- **Scalability limitations**
  Many networks struggle with throughput and high transaction fees during peak demand.
- **Security and custody risks**
  Users are responsible for private keys, and losses are often irreversible.

### *Disruptive potential*

- **Challenge to traditional banking**
  Crypto undermines legacy payment rails, remittances, and settlement systems.
- **New financial instruments**
  Enables decentralised finance (DeFi), tokenisation, and programmable money.
- **Pressure on regulators and institutions**
  Forces adaptation of legal, monetary, and compliance frameworks.

**In summary**, cryptocurrencies are not merely speculative assets. They represent a structural challenge to traditional financial markets by redefining trust, ownership, and value transfer, though widespread adoption depends on resolving volatility, regulation, and scalability issues.

Question 5 – Future of Blockchain [25 Marks]

A) There is a wealth of critical discussion based upon the disruptive nature of Crypto Currencies on established economies and markets, but with this comes the tremendous potential for the underlying Blockchain technology. Provide 3 use cases where blockchain technology could benefit existing markets and processes, critically discuss in detail the advantages and disadvantages for each use case mentioned. [15 Marks]

## ANSWER: Blockchain use cases beyond cryptocurrencies: benefits and limitations (15 marks)

While cryptocurrencies dominate public debate, the real long-term value lies in **blockchain as an enabling infrastructure**. Below are **three concrete use cases** where blockchain can benefit existing markets and processes, along with a critical discussion of advantages and disadvantages.

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## 1. Supply Chain Management and Provenance Tracking

**Use case**
Blockchain can be used to track goods from origin to consumer, recording every handoff immutably. This is already explored using platforms such as **Hyperledger Fabric**.

**Advantages**

- End-to-end transparency and traceability
- Reduced fraud, counterfeiting, and disputes
- Improved trust between untrusted parties

**Disadvantages**

- "Garbage in, garbage out": blockchain cannot verify false data at entry
- Integration with legacy systems is complex and costly
- Limited incentive for all supply chain actors to participate honestly

**Critical insight**
Blockchain improves trust *between organisations*, not necessarily data accuracy itself.

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## 2. Financial Settlement and Cross-Border Payments

**Use case**
Blockchain can be used for near-instant settlement of cross-border payments and asset transfers, reducing reliance on slow intermediaries such as correspondent banks.

**Advantages**

- Faster settlement times (minutes instead of days)
- Lower transaction and reconciliation costs
- Reduced counterparty and settlement risk

**Disadvantages**

- Regulatory and compliance challenges across jurisdictions
- Volatility in public blockchain assets
- Scalability concerns during peak demand

**Critical insight**
Blockchain is well suited for back-end settlement layers, but full replacement of traditional rails remains unlikely in the short term.

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## 3. Digital Identity and Identity Management

**Use case**
Blockchain-based digital identity systems allow individuals to control and selectively disclose identity credentials, aligning with self-sovereign identity models.

**Advantages**

- User ownership and control of identity data
- Reduced identity fraud and duplication
- Streamlined verification processes

**Disadvantages**

- Key management complexity for users
- Privacy concerns if poorly implemented
- Legal recognition varies across regions

**Critical insight**
Blockchain improves *identity portability and control*, but governance and usability remain major barriers to adoption.

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## Conclusion

What this really means is that blockchain's value lies not in replacing entire systems overnight, but in **optimising trust, coordination, and transparency where multiple parties interact without full trust**. Each use case offers clear advantages, but also exposes trade-offs in cost, complexity, and governance. The future of blockchain adoption will depend less on technical capability and more on **practical integration, regulation, and incentive alignment**.

B) A basic smart contract can be thought of as similar to a vending machine. Anyone who wants to interact with the machine can check the price of its goods. If they agree with the price, they can place the necessary payment in the machine and receive the goods. The machine takes payment, compares the amount received with the displayed price to figure out if there are enough funds to pay for the good. If the amount paid is higher than the displayed price they must also calculate the change due and return the change. This is all done without human intervention. Smart contract go beyond the vending machine proposal in proposing to embed contracts in all sorts of property that is valuable and controlled by digital means. Szabo, 1997 A List of potential use cases for smart contracts can be seen below. Select two use cases from the list, and critically evaluate the use of smart contracts in relation to the implementation of those use cases

i. Basic Financial transactions

ii. Complex financial derivatives calculations e.g. futures, forwards options

iii. Governance

iv. Crowd Sourcing

v. Raising capital

vi. Prediction Markets [10 Marks]

ANSWER: Smart contracts, originally proposed by **Nick Szabo**, automate agreements using code. Below is a **concise but critical evaluation of all six listed use cases**, focusing on strengths, limitations, and suitability.

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## i. Basic Financial Transactions

- **Advantages**
  - Automated payments and escrow without intermediaries
  - Faster settlement and lower transaction costs
  - High transparency and auditability
- **Disadvantages**
  - Irreversible execution if errors occur
  - Code bugs can lead to permanent fund loss
- **Assessment**
  Highly suitable due to deterministic logic and minimal ambiguity.

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## ii. Complex Financial Derivatives (futures, options, forwards)

- **Advantages**

    - Automated execution of complex financial logic
    - Reduced counterparty risk
    - Increased transparency of contract terms
- **Disadvantages**
    - Reliance on external data (oracles) introduces trust assumptions
    - High complexity increases risk of coding errors
- **Assessment**
  Powerful but risky; correctness and oracle reliability are critical.

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### iii. Governance

- **Advantages**
    - Transparent voting and rule enforcement
    - Reduced corruption and manipulation
    - Enables decentralised autonomous organisations (DAOs)
- **Disadvantages**
    - Voter apathy and governance attacks
    - Code-based rules can be difficult to amend
- **Assessment**
  Effective for procedural governance, weak for nuanced decision-making.

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### iv. Crowd Sourcing

- **Advantages**
    - Automatic fund collection and conditional release
    - Reduces reliance on trusted platforms
    - Transparent use of funds
- **Disadvantages**
    - Limited protection for contributors
    - Difficult to handle failed or disputed outcomes
- **Assessment**
  Suitable for clearly scoped projects with measurable outcomes.

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### v. Raising Capital (ICOs / Token Sales)

- **Advantages**
    - Global, permissionless fundraising
    - Automated token distribution
    - Lower barriers to entry
- **Disadvantages**

    - High fraud risk
    - Weak investor protection
    - Regulatory uncertainty
- **Assessment**
  Disruptive but dangerous without governance and regulation.

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## vi. Prediction Markets

- **Advantages**
    - Market-driven forecasting
    - Reduced central control
    - Incentivises truthful information
- **Disadvantages**
    - Oracle manipulation risk
    - Ethical and legal concerns
- **Assessment**
  Technically strong but socially and legally constrained.

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## Overall conclusion (short)

Smart contracts work best where **rules are explicit, outcomes are measurable, and discretion is minimal**. As complexity and reliance on external data increase, so do risks. Their disruptive potential is real, but success depends on **careful use-case selection, strong governance, and secure implementation**.

# 2023-24-Paper

## Question 1 (5 marks)

**Public Ethereum network consensus mechanism:**
✅ **3. POS (Proof of Stake)**

## Question 2 (5 marks)

**Central novelty of the Bitcoin blockchain:**
✅ **3. The use of Proof Of Work to provide decentralized consensus**

## Question 3 (5 marks)

**Properties of money (select all that apply):**
✅ **1. Medium of Exchange**
✅ **3. Unit of Account**
✅ **5. Store of Value**
✅ **7. Resistance to Counterfeiting**

## Question 4 (5 marks)

**A blockchain hard fork is caused by:**
✅ **3. A software change invalidating future blocks generated by previous software version**

## Question 5 (5 marks)

**Purpose of gas in Ethereum:**
✅ **2. To prevent infinite loops in smart contract code**

## Question 6 (5 marks)

**Bitcoin transactions are:**
✅ **2. Pseudonymous and traceable**

## Question 7 (5 marks)

**Ethereum block generation time:**
✅ **2. 15 seconds**

Question 8: (5 Marks) What is Proof of Stake? Detail your response including references to randomness, its role in decentralized consensus, power consumption and Delegated Proof of Stake.

Answer: **Proof of Stake (PoS)** is a blockchain consensus mechanism where **validators are selected to create and validate blocks based on the amount of cryptocurrency they stake**, rather than computational power.

- **Randomness**
  Validators are chosen using pseudo-random selection mechanisms, often weighted by stake size, to prevent predictable control and reduce centralisation.
- **Role in decentralised consensus**
  PoS enables distributed nodes to agree on the ledger state without relying on energy-intensive mining, maintaining decentralisation through economic incentives and penalties (slashing).
- **Power consumption**
  PoS dramatically reduces energy usage compared to Proof of Work, as it does not require continuous hashing or specialised hardware.
- **Delegated Proof of Stake (DPoS)**
  In DPoS, token holders vote to elect a limited number of delegates who validate blocks on their behalf, improving scalability but reducing decentralisation.

In summary, Proof of Stake replaces computational cost with **economic risk**, providing a more energy-efficient approach to decentralised consensus.

Question 9 (20 marks) Define the four fundamental concepts, which you believe made the creation of the bitcoin blockchain possible in 2009, and how these concepts interact and combine to enable blockchain as a disruptive technology. You must also critically appraise why you have selected these as the most fundamental concepts.

ANSWER: **Four fundamental concepts enabling the creation of Bitcoin and blockchain as a disruptive technology**

Bitcoin's release in 2009 by **Satoshi Nakamoto** was not the result of a single breakthrough, but the **careful integration of several foundational concepts**. Many technologies existed before Bitcoin, but four concepts are truly fundamental because **removing any one of them would prevent blockchain from functioning as intended**.

## 1. Cryptography (Hashing and Digital Signatures)

**Definition**
Cryptography provides secure mathematical mechanisms for data integrity, ownership, and authentication. In Bitcoin, this primarily includes cryptographic hash functions and public–private key cryptography.

**Role in Bitcoin**

- Hashing ensures immutability and tamper detection
- Digital signatures prove ownership of funds
- Private keys authorise transactions without revealing identity

**Critical appraisal**
Cryptography is essential because blockchain trust is **mathematical rather than institutional**. Without cryptography, there is no secure ownership or data integrity. However, cryptography alone does not solve coordination or agreement across a network, which limits its standalone usefulness.

## 2. Distributed Ledger Technology (DLT)

**Definition**
A distributed ledger is a shared database replicated across many independent nodes, where no single party has control.

**Role in Bitcoin**

- Every node maintains a full copy of the transaction history
- Eliminates single points of failure
- Enables transparency and auditability

**Critical appraisal**
DLT removes the need for a central authority, which is core to Bitcoin's disruption of traditional finance. However, a distributed ledger without a coordination mechanism quickly fragments. This makes DLT necessary but insufficient on its own.

## 3. Consensus Mechanisms (Proof of Work)

**Definition**
A consensus mechanism allows decentralised participants to agree on a single valid state of the ledger, even in the presence of malicious actors.

**Role in Bitcoin**

- Proof of Work determines which block is added next
- Prevents double spending
- Makes attacks economically infeasible

**Critical appraisal**
Consensus is the key innovation that allows decentralised trust. Proof of Work is energy-intensive, but it is extremely robust and battle-tested. Without consensus, a distributed ledger has no authoritative version of history.

## 4. Economic Incentives and Game Theory

**Definition**
Economic incentives align rational self-interest with honest network participation.

**Role in Bitcoin**

- Miners are rewarded with newly minted Bitcoin and fees
- Dishonest behaviour is economically punished
- Network security becomes self-sustaining

**Critical appraisal**
This concept is often underestimated, but it is what keeps the system alive. Cryptography and consensus define the rules; incentives ensure those rules are followed. Poorly designed incentives can cause centralisation, but without incentives, decentralised systems fail due to apathy or attack.

### How these concepts combine to enable disruption

What this really means is that Bitcoin works because:

- **Cryptography** secures ownership and data
- **Distributed ledgers** remove central control
- **Consensus mechanisms** create agreement without trust
- **Economic incentives** enforce honest behaviour

Individually, these concepts existed before 2009. Combined, they created the first **trustless, decentralised monetary system**, capable of operating globally without intermediaries.

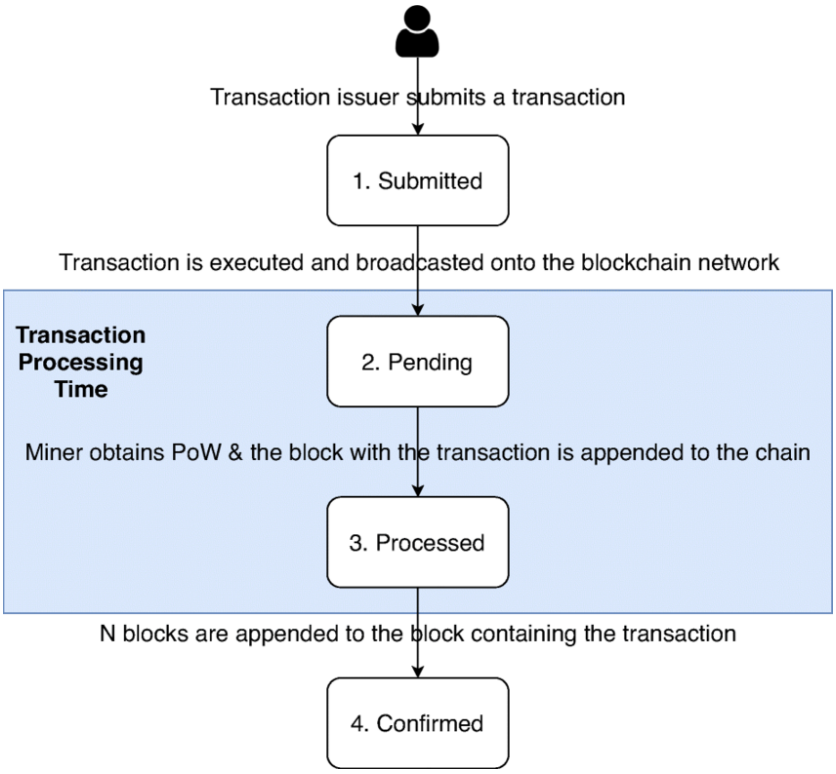
### Why these are the most fundamental concepts

### These four were selected because they are:

- **Necessary**: removing any one breaks the system
- **Complementary**: each solves a different core problem
- **General**: applicable across public, private, and hybrid blockchains

Question 10 (10 marks) Critically explain and detail a transaction lifecycle process, you are free to choose either Bitcoin or Ethereum transaction lifecycle. Within your response ensure to detail each key function and process within the lifecycle, an illustration or diagram can be used to support your response.

ANSWER:




## 1. Transaction creation

- A user creates a transaction using a wallet (e.g. MetaMask).
- The transaction includes:
  - Sender and receiver address
  - Amount of ETH or smart contract interaction
  - Gas limit and gas price
- The transaction is **digitally signed** with the sender's private key.

## 2. Transaction broadcast

- The signed transaction is broadcast to the Ethereum peer-to-peer network.
- It enters the **mempool**, where it waits to be processed.

## 3. Initial validation by nodes

- Nodes check that:
  - The digital signature is valid
  - The sender has enough ETH to cover value + gas
  - The nonce is correct
- Invalid transactions are rejected.

## 4. Block proposal by validators (Proof of Stake)

- A validator is **randomly selected** to propose the next block.
- The validator chooses transactions from the mempool, usually prioritising higher gas fees.

## 5. Smart contract execution

- If the transaction calls a smart contract:
  - Code is executed on the Ethereum Virtual Machine (EVM)
  - Gas is consumed for each operation
- If gas runs out, execution stops but gas is still charged.

## 6. Block confirmation and finalisation

- Other validators verify the proposed block.
- Once agreed, the block is added to the blockchain.
- The transaction now has **confirmations** and becomes increasingly difficult to reverse.

## 7. State update

- Account balances are updated.
- Smart contract state changes are recorded.
- Gas fees are paid to the validator.

## Critical insight

Ethereum's transaction lifecycle is more complex than Bitcoin's because it supports **programmable smart contracts**, not just value transfers. This adds flexibility and power, but also introduces risks such as:

- Smart contract bugs
- Higher fees during congestion
- Execution failures due to gas limits

## Summary

The Ethereum transaction lifecycle combines:

- Digital signatures for security
- Gas for execution control
- Proof of Stake for consensus
- Smart contracts for programmability

Together, these steps allow Ethereum to function as a **decentralised global computing platform**, not just a digital currency network.

Question 11 (5 marks) Detail why you are expected to wait for a certain number of confirmations after receiving a cryptocurrency transaction?

ANSWER: **Why confirmations are required after a cryptocurrency transaction**

- **Protection against double spending**
  Waiting for confirmations ensures the sender cannot spend the same coins again in another transaction.
- **Chain reorganisation risk**
  Early blocks can sometimes be replaced. More confirmations reduce the chance of a transaction being reversed.
- **Increased security**
  Each new block added after the transaction makes it harder and more expensive to alter the transaction.
- **Network consensus**
  Confirmations show that the majority of the network has accepted the transaction as valid.
- **Finality assurance**
  After a sufficient number of confirmations, the transaction is considered final and irreversible.

**In short**, confirmations provide confidence that a transaction is permanently recorded on the blockchain and cannot be undone.

Question 12 (10 marks) The European Union in April of 2023 has passed legislation relating to the mining and holding of Crypto-Assets. Provide a brief summary of this legislation.

ANSWER: **EU legislation on crypto-assets (April 2023 – MiCA)**

In April 2023, the European Union passed the **Markets in Crypto-Assets Regulation (MiCA)**, which is the first comprehensive legal framework for crypto-assets across all EU member states.

- **Purpose of the legislation**
  MiCA aims to regulate the issuance, trading, and holding of crypto-assets to provide legal clarity, protect consumers, and reduce financial risk.
- **Regulation of crypto service providers**
  Crypto exchanges, wallet providers, and issuers must be licensed and supervised within the EU and comply with strict operational rules.
- **Consumer protection**
  Firms must provide clear information on risks, prevent misleading advertising, and safeguard customer funds.
- **Stablecoin rules**
  Issuers of stablecoins must hold sufficient reserves and meet transparency requirements to reduce systemic risk.
- **Anti-money laundering (AML) compliance**
  MiCA strengthens AML and counter-terrorist financing rules, including transaction monitoring and identity checks.
- **Environmental considerations**
  The legislation addresses concerns around energy-intensive mining by requiring transparency on environmental impact, though Proof of Work was not banned.

**In summary**, MiCA brings crypto-assets into the EU's regulated financial system, increasing trust and stability while potentially raising compliance costs and reducing anonymity for users.

Question 13 (5 Marks) Explain the concept of an ICO, within your response highlight a positive and negative impact of such an action.

ANSWER: **Initial Coin Offering (ICO)**

An **Initial Coin Offering (ICO)** is a fundraising method where a blockchain project raises capital by selling newly created digital tokens to investors, usually in exchange for cryptocurrencies such as Bitcoin or Ethereum.

- **Positive impact**
  ICOs allow startups to raise funds quickly and globally without relying on traditional banks or venture capital, encouraging innovation and open participation.
- **Negative impact**
  ICOs carry high risk due to limited regulation, which has led to scams, poor-quality projects, and weak investor protection.

**In summary**, ICOs are a powerful but risky funding mechanism that can drive innovation while also exposing investors to significant financial risk.

Question 14 (10 marks) Briefly Explain the concept of POW Mining in the BTC network.

ANSWER: **Proof of Work (PoW) mining** is the method used by the **Bitcoin network** to add new blocks to the blockchain and keep the system secure without a central authority.

- **Collecting transactions**
  Miners collect new and unconfirmed Bitcoin transactions from the network.
- **Solving a puzzle**
  Miners compete to solve a mathematical puzzle by repeatedly hashing the block data. This requires a lot of computing power.
- **Finding a valid block**
  The first miner to find a correct solution creates a new block and shares it with the network.
- **Verification by other nodes**
  Other computers on the network check the solution and the transactions to make sure everything follows the rules.
- **Block reward**
  The miner who creates the block is rewarded with newly created Bitcoins and transaction fees.
- **Security of the network**
  Because solving the puzzle is difficult and costly, it is very hard for attackers to change past transactions or cheat the system.

**In short**, Proof of Work uses computing effort to secure the Bitcoin network and allows everyone to agree on a single, trusted version of the blockchain without needing a bank or central authority.